Experiment – 4 Protocols Learnings in detail using Wireshark filter and flow Graph

Aim

To capture network packets using **Wireshark** and identify various network protocols using **filters and flow graph**.

Procedure

1. Open **Wireshark** and select the active network interface.
2. Start **packet capture**.
3. Generate network traffic (browse websites or ping).
4. Stop the capture after a few seconds.
5. Apply filters such as:
   - `tcp`
   - `udp`
   - `arp`
   - `icmp`
   - `dns`
6. Analyze packet details like **source IP, destination IP, port numbers, and protocol type**.
7. Use **Statistics → Flow Graph** to view packet communication.

Protocols Observed

- TCP
- UDP
- ARP
- ICMP
- DNS

Protocol Explanation

1. UDP

Filter: `udp`
UDP is a **connectionless protocol** used for fast data transmission without acknowledgements.
Example uses: **Streaming, gaming, DNS queries**.

2. TCP

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Filter: `tcp`
TCP is a **connection-oriented protocol** that provides reliable communication using the **three-way handshake**.
Example uses: **Web browsing, email, file transfer**.

3. ARP

Filter: `arp`
ARP converts **IP address to MAC address** in a local network.

4. ICMP

Filter: `icmp`
ICMP is used for **network diagnostics** such as the **ping command**.

5. DNS

Filter: `dns`

DNS converts **domain names into IP addresses**.

Result

Thus, network packets were captured and analyzed using **Wireshark**, and various protocols such as **TCP, UDP, ARP, ICMP, and DNS** were identified using filters and flow graph